

Simple Linear Regression:

Simple Linear Regression is used to predict the dependent variable (Y) based on one independent variable (X).

The equation of simple linear regression is:

$$\hat{Y} = a + bX$$

Where:

- a = Intercept
- b = Regression coefficient (slope)
- X = Independent variable
- $\hat{Y}$ = Predicted value of Y

The formulas are:

$$b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$
$$a = \frac{\sum Y - b \sum X}{n}$$

Problem 1:

The following table shows the number of hours students studied and the marks obtained in an examination. Calculate the regression equation and predict the marks for a student who studies for 6 hours.

Student	Study (X)	Hours	Marks (Y)
1	1		35
2	2		40

Student	Study Hours (X)	Marks (Y)
3	3	50
4	4	55
5	5	65

Solution:

X	Y	X ²	XY
1	35	1	35
2	40	4	80
3	50	9	150
4	55	16	220
5	65	25	325
15	245	55	810

$$n = 5$$

$$\sum X = 15, \quad \sum Y = 245$$

$$\sum X^2 = 55, \quad \sum XY = 810$$

Calculate the Slope (b)

$$b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

Substitute the values:

$$b = \frac{5(810) - 15(245)}{5(55) - 15^2}$$

$$b = \frac{4050 - 3675}{275 - 225}$$

$$b = \frac{375}{50}$$

$$b = 7.5$$

Calculate the Intercept (a)

$$a = \frac{\sum Y - b \sum X}{n}$$

$$a = \frac{245 - 7.5(15)}{5}$$

$$a = \frac{245 - 112.5}{5}$$

$$a = 26.5$$

Regression Equation

$$\hat{Y} = a + bX$$

$$\hat{Y} = 26.5 + 7.5X$$

Predict Marks for 6 Hours

$$X = 6$$

$$\hat{Y} = 26.5 + 7.5(6)$$

$$\hat{Y} = 26.5 + 45$$

$$\hat{Y} = 71.5$$

For every additional hour of study, the predicted examination marks increase by approximately 7.5 marks.

Problem 2:

The following table shows the advertising expenditure and corresponding sales of a company. Calculate the regression equation and predict sales when the advertising expenditure is ₹12,000.

Month	Advertising Expenditure (X)	Sales (Y)
1	2	20
2	4	25
3	6	35
4	8	40
5	10	50

Solution:

Construct the Calculation Table

X	Y	X²	XY
2	20	4	40
4	25	16	100
6	35	36	210
8	40	64	320
10	50	100	500
30	170	220	1170

$$n = 5$$

$$\sum X = 30, \quad \sum Y = 170$$

$$\sum X^2 = 220, \quad \sum XY = 1170$$

Calculate the Slope (b)

$$b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

$$b = \frac{5(1170) - 30(170)}{5(220) - 30^2}$$

$$b = \frac{5850 - 5100}{1100 - 900}$$

$$b = \frac{750}{200}$$

$$b = 3.75$$

Calculate the Intercept (a)

$$a = \frac{\sum Y - b \sum X}{n}$$

$$a = \frac{170 - 3.75(30)}{5}$$

$$a = \frac{170 - 112.5}{5}$$

$$a = 11.5$$

Regression Equation

$$\hat{Y} = a + bX$$

$$\hat{Y} = 11.5 + 3.75X$$

Predict Sales for ₹12,000 Advertising Expenditure

Since X is measured in thousands of rupees:

$$X = 12$$

$$\hat{Y} = 11.5 + 3.75(12)$$

$$\hat{Y} = 11.5 + 45$$

$$\hat{Y} = 56.5$$

Multiple linear regression

Multiple Linear Regression is used to predict a dependent variable (Y) using two or more independent variables.

$$\hat{Y} = a + b_1X_1 + b_2X_2$$

Where:

- a = Intercept
- b_1, b_2 = Regression coefficients
- X_1, X_2 = Independent variables
- $\hat{Y}$ = Predicted value

Problem1:

The following table shows the study hours, attendance percentage and examination marks of students. Calculate the multiple linear regression equation and predict the marks for a student who studies for 6 hours and has 85% attendance.

Student	Study Hours (X ₁)	Attendance (X ₂)	Marks (Y)
1	1	60	65
2	2	70	80
3	3	60	75
4	4	80	100
5	5	70	95

Solution:

Regression Equation

$$\hat{Y} = a + b_1 X_1 + b_2 X_2$$

The coefficients are calculated using the **least squares method**.

$$\begin{aligned}\sum Y &= na + b_1 \sum X_1 + b_2 \sum X_2 \\ \sum X_1 Y &= a \sum X_1 + b_1 \sum X_1^2 + b_2 \sum X_1 X_2 \\ \sum X_2 Y &= a \sum X_2 + b_1 \sum X_1 X_2 + b_2 \sum X_2^2\end{aligned}$$

Substituting

$$415 = 5a + b_1 \sum X_1 + b_2 \sum X_2$$

$$1325 = 5a + b_1 \sum X_1^2 + b_2 \sum X_1 X_2$$

$$28650 = 5a + b_1 \sum X_1 X_2 + b_2 \sum X_2^2$$

X ₁	X ₂	Y	X ₁ ²	X ₂ ²	X ₁ X ₂	X ₁ Y	X ₂ Y
1	60	65	1	3600	60	65	3900
2	70	80	4	4900	140	160	5600
3	60	75	9	3600	180	225	4500
4	80	100	16	6400	320	400	8000
5	70	95	25	4900	350	475	6650
15	340	415	55	23400	1050	1325	28650

the calculated sums:

Solving these equations:

$$a = 35, \quad b_1 = 5, \quad b_2 = 0.5$$

Regression Equation

$$\hat{Y} = 35 + 5X_1 + 0.5X_2$$

Prediction

- Study Hours = 6
- Attendance = 85%

$$\begin{aligned}\hat{Y} &= 35 + 5(6) + 0.5(85) \\ &= 35 + 30 + 42.5 \\ \hat{Y} &= 107.5\end{aligned}$$

The model estimates the marks based on study hours and attendance. The predicted value is an extrapolation beyond the observed marks range, so it should be interpreted cautiously.

Problem 2:

The following table shows the advertising expenditure, social media promotion expenditure and sales of a company. Calculate the multiple linear regression equation and predict sales when advertising expenditure is ₹6,000 and social media expenditure is ₹4,000. Assume all expenditure and sales values are measured in thousands of rupees.

Month	Advertising (X_1)	Social Media (X_2)	Sales (Y)
1	1	1	17
2	2	1	19
3	1	2	20
4	3	2	23

Month	Advertising (X_1)	Social Media (X_2)	Sales (Y)
5	2	3	24
6	4	3	27

Solution:

X_1	X_2	Y	X_1^2	X_2^2	X_1X_2	X_1Y	X_2Y
1	1	17	1	1	1	17	17
2	1	19	4	1	2	38	19
1	2	20	1	4	2	20	40
3	2	23	9	4	6	69	46
2	3	24	4	9	6	48	72
4	3	27	16	9	12	108	81
13	12	130	35	28	29	300	275

$$130 = 6a + 13b_1 + 12b_2$$

$$300 = 13a + 35b_1 + 29b_2$$

$$275 = 12a + 29b_1 + 28b_2$$

$$a = 10, \quad b_1 = 2, \quad b_2 = 3$$

Regression Equation

$$\hat{Y} = 10 + 2X_1 + 3X_2$$

Advertising expenditure = ₹6,000 → $X_1=6$

Social media expenditure = ₹4,000 → $X_2=4$

$$\begin{aligned}\hat{Y} &= 10 + 2(6) + 3(4) \\ &= 10 + 12 + 12\end{aligned}$$

$$\hat{Y} = 34$$

Since Y is measured in thousands of rupees:

Predicted Sales = ₹34,000

The model estimates sales using both advertising expenditure and social media expenditure.

RIDGE REGRESSION

Ridge Regression is a regularized version of linear regression that is used to reduce overfitting and handle multicollinearity among input features. It adds a penalty term (L2 regularization) to the ordinary least squares cost function.

Mathematical Equation

The linear regression equation is:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

Ridge Regression minimizes:

$$J(\beta) = \sum_{i=1}^m (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^n \beta_j^2$$

Symbol	Meaning
y_i	Actual output
$\hat{y}_i$	Predicted output
β_j	Model coefficients
λ	Regularization parameter
m	Number of samples
n	Number of features

Effect of Regularization

Lambda (λ)	Effect
0	Equivalent to ordinary linear regression
Small value	Slightly reduces coefficient magnitudes
Large value	Strongly shrinks coefficients
Very large value	May cause underfitting

Ridge regression shrinks coefficients toward zero but generally does not make them exactly zero.

Feature	Linear Regression	Ridge Regression
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Feature	Linear Regression	Ridge Regression
Regularization	No	Yes (L2)
Cost function	Sum of squared errors	SSE + penalty
Overfitting	Can occur	Can be reduced
Multicollinearity	Can cause unstable coefficients	Helps stabilize coefficients
Coefficients	Unrestricted by penalty	Shrunk toward zero
Feature selection	No inherent feature selection	No inherent exact feature selection
Complexity	Simple	Includes hyperparameter tuning

Problem1:

A real-estate company wants to predict house prices based on house area. Using the given dataset, calculate the regression equation using (a) Linear Regression and (b) Ridge Regression with $\lambda = 1$. Predict the price of a 500 sq.ft. house and compare the two models. Explain why the Ridge coefficient differs from the Linear Regression coefficient.

The following data are collected:

House	Area X (100 sq.ft.)	Price Y (₹ lakh)
1	1	2
2	2	4
3	3	5

House	Area X (100 sq.ft.)	Price Y (₹ lakh)
4	4	8

Solution:

Prepare the data

Area X	Price Y	X^2	XY
1	2	1	2
2	4	4	8
3	5	9	15
4	8	16	32
$\Sigma = 10$	$\Sigma = 19$	$\Sigma = 30$	$\Sigma = 57$

Linear Regression:

intercept is:

$$\hat{Y} = bX$$

$$b = \frac{57}{30}$$

$$\hat{Y} = 1.9X$$

coefficient is:

$$b = \frac{\sum XY}{\sum X^2}$$

$$b = 1.9$$

Prediction for 500 sq.ft.

Since X is measured in units of 100 sq.ft.:

$$500 \text{ sq.ft.} = 5$$

$$\hat{Y} = 1.9(5)$$

$$\hat{Y} = 9.5 \text{ lakh}$$

Ridge Regression

$$\lambda = 1$$

The Ridge coefficient is:

$$b_R = \frac{\sum XY}{\sum X^2 + \lambda}$$

$$b_R = \frac{57}{30 + 1}$$

$$b_R = \frac{57}{31}$$

$$b_R = 1.8387$$

The Ridge equation is:

$$\hat{Y} = 1.8387(5)$$

$$\hat{Y} = 9.1935 \text{ lakh}$$

Comparison:

Parameter	Linear Regression	Ridge Regression
Application	House price prediction	House price prediction
λ	0	1

Parameter	Linear Regression	Ridge Regression
$\sum X^2$	30	30
$\sum XY$	57	57
Coefficient	1.9000	1.8387
Regression equation	$\hat{Y} = 1.9X$	$\hat{Y} = 1.8387X$
Area	500 sq.ft.	500 sq.ft.
Predicted price	₹9.50 lakh	₹9.1935 lakh
Coefficient magnitude	Higher	Lower
Regularization	No	Yes
Effect on coefficient	No shrinkage	Shrinks toward zero

For the same house-price dataset:

Linear coefficient = 1.9

Ridge coefficient = 1.8387

The Ridge coefficient is smaller because Ridge adds an **L2 penalty**:

$$\text{Ridge Cost} = \sum (Y - \hat{Y})^2 + \lambda b^2$$

- **Linear Regression:** predicts **₹9.50 lakh**
- **Ridge Regression:** predicts **₹9.19 lakh**

Difference:

$$9.50 - 9.1935$$

₹0.3065 lakh

approximately **₹30,650**.

- The important point is not that Ridge always gives a lower prediction. The important point is that:
- Ridge Regression reduces the magnitude of regression coefficients by adding an L2 regularization penalty.
- This becomes particularly useful in real applications such as house-price prediction, sales forecasting, medical prediction, and energy-demand prediction, where several input variables may be correlated.

Problem2:

A retail company wants to predict daily sales based on advertising expenditure. The following data are collected:

Advertising Expenditure (X)	Sales (Y)
1	2
2	3
3	5
4	6

Assume the regression line passes through the origin.

- a) Calculate the regression equation using **Linear Regression**.
- b) Calculate the regression equation using **Ridge Regression** with $\lambda = 1$.
- c) Predict sales when advertising expenditure is 5 units using both models. Compare the regression coefficients and predicted values.

Solution:

Parameter	Linear Regression	Ridge Regression
Application	Sales prediction	Sales prediction
Regularization (λ)	0	1
$\sum XY$	47	47
$\sum X^2$	30	30
Coefficient	1.5667	1.5161
Regression equation	$\hat{Y} = 1.5667X$	$\hat{Y} = 1.5161X$
Advertising expenditure	5	5
Predicted sales	7.8335	7.5805
Coefficient magnitude	Higher	Lower

Problem3:

An electricity company wants to predict daily electricity consumption based on the number of operating industrial machines. The following data are collected:

Number of Machines (X)	Electricity Consumption (Y)
1	3
2	8
3	15
4	20

Assume that the regression line passes through the origin.

- a) Calculate the regression equation using Linear Regression.
- b) Calculate the regression equation using Ridge Regression with $\lambda = 20$.
- c) Predict electricity consumption when the number of machines is 5 using both models. Compare the slopes and predictions.

Solution:

Parameter	Linear Regression	Ridge Regression
Application	Electricity consumption	Electricity consumption
Regularization (λ)	0	20
$\sum XY$	144	144
$\sum X^2$	30	30
Slope	4.80	2.88
Regression equation	$\hat{Y} = 4.8X$	$\hat{Y} = 2.88X$
Number of machines	5	5
Predicted consumption	24	14.4
Slope reduction	—	40%
Regularization	No	L2 penalty